AUS | الجامعة الأمريكية في الشارقة
American University of Sharjah

MTH 343

Numerical Analysis

Fall 2024

Assignment #1
Section #3
Due on 29^{th} 2024

Title: Muller's Method Algorithm

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(a) Muller Method M-file function

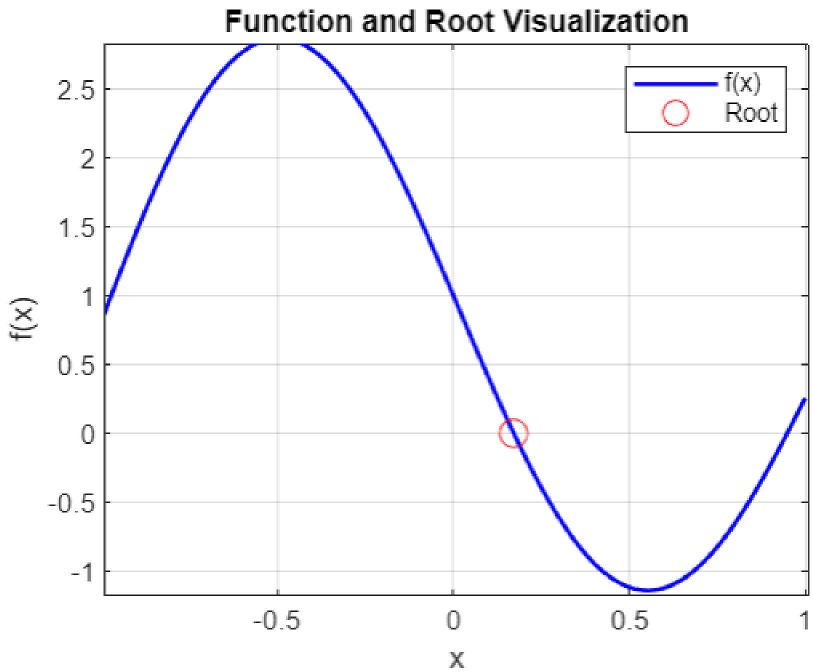
```
function table = muller_method(f, p0, p1, p2, tol, maxIter)
    table = [];
    i = 0;

    while i < maxIter
        i = i + 1;

        % Equations taken and implemented from the Numerical Analysis
        % notebook (p3, c, b, a)
        c = f(p2);

        b = ((p0-p2)^2 * (f(p1) - f(p2)) - (p1-p2)^2 * (f(p0) - f(p2))) / ((p0-
p2)*(p1-p2)*(p0-p1));

        a = ((p1-p2) * (f(p0) - f(p2)) - (p0-p2) * (f(p1) - f(p2))) / ((p0-p2)*(p1-
p2)*(p0-p1));

        discriminant = b^2 - 4 * a * c;
        if discriminant < 0
            error("Error: Complex root encountered.");
        end

        sqrtDisc = sqrt(discriminant);
        p3 = p2 - (2 * c) / (b + sign(b) * sqrtDisc);

        relativeError = abs((p3 - p2) / p3);

        table = [table; i, p3, relativeError];

        if relativeError <= tol
            break;
        end

        % This part updates the values for the next iteration
        p0 = p1;
        p1 = p2;
        p2 = p3;
    end
end
```

(b) Main M-file

```matlab
f = @(x) cos(x) - 2*sin(3*x);

p0 = 0;
p1 = 0.25;
p2 = 0.5;

tol = 10^-6;
maxIter = 100;

% Calling the muller method from the other M-file with the function
table = muller_method(f, p0, p1, p2, tol, maxIter);

% Plotting the function
x = linspace(-1, 1, 100);
plot(x, f(x), "b-", "LineWidth", 1.5);
hold on;

% Plotting the root
plot(table(end, 2), f(table(end, 2)), "ro", "MarkerSize", 10);
hold off;

title("Function and Root Visualization");
xlabel("x");
ylabel("f(x)");
grid on;
legend("f(x)", "Root")
xlim([-1 1]);
ylim([-2 2]);
hold off;

% Displaying the table and it's values
fprintf("# of iteration\t| Value of the Root (pn) \t| Approximate Relative Error\n");
fprintf("%.0f\t\t%.5f\t\t\t\t%.5e\n", table');
```

Output: -

```
>> main
# of iteration  | Value of the Root (pn)        | Approximate Relative Error
1               0.16585                         2.01468e+00
2               0.17154                         3.31609e-02
3               0.17171                         9.70571e-04
4               0.17171                         2.23903e-06
5               0.17171                         3.18533e-12
>>
```


Function and Root Visualization
f(x)
Root
2.5
2
1.5
1
0.5
0
-0.5
-1
-0.5
0
0.5
1
x
f(x)